Table 6.2: notation of the leak codes

code	meaning	evidence
M	identical variant row in training (same protein family, identical mutation set)	verified
m	every mutation in the row appears in training in the same protein, but not this combination (e.g. a test double whose two singles were trained)	verified
P	same protein (≥95 % identity over ≥90 % coverage, or ≥90 % identity over ≥50 %) in training, no mutation match	verified
H	strong homolog in training: MMseqs2 E ≤ 1e-3 over ≥50 % of the test protein	verified
h	weak/ambiguous: E ≤ 1e-2 or ≥30 % identity over ≥30 % of the test protein	measured
D	documented training on the same source dataset; rows not reconstructable	documented
C	pretraining-corpus exposure only (WT sequence/structure, no mutation supervision)	documented
U	training data not disclosed	documented
V	the dataset was used for validation or model selection	documented
#	the number is not a test result (the dataset is the training set)	verified
?	probable but unverified (e.g. M?: same protein and same source dataset, row-level file not public)	inferred
n.a.	not audited: no statement about overlap is made, and the cell is not eligible for bold	–

Table 6.3: training inventories

inventory	training data	source	evidence
ESM-MSR (released), ESM-MSR (devel), σ sweep, WT/MT heads	all substitutions of the 172 capped-train libraries (229,721 rows)	splits_oct06_capped.pkl + Tsuboyama2023 _Dataset2_Dataset3_20230416.csv	verified
MutateEverything st1	the 172 capped-train libraries: 154,358 valid singles (150,815 wild-type-backbone + 3,543 mutant-background)	capped split; ME README recipe, stage 1 (20 epochs)	verified
MutateEverything st2	the same singles plus all 53,188 measured doubles (207,546 rows); stage 2 (100 epochs) initialised from the same seed's stage 1	capped split; ME README recipe	verified
ThermoMPNN (additive)	the same 154,358 singles	capped split; train_single.yaml	verified
ThermoMPNN-D	the 53,188 measured doubles plus 118,963 augmented doubles (upstream double-aug, built from the singles that have a Rosetta model: 79 % of them)	capped split; train_epistatic.yaml	verified
SPURS (retrained)	the 172 capped-train libraries: 154,358 valid singles (150,815 wild-type-backbone + 3,543 mutant-background)	capped split; patched SPURS training code, three seeds	verified
SPURS multi (retrained)	the same singles (through its same-seed SPURS initialisation) and the 53,188 measured doubles	capped split; patched SPURS training code, three seeds	verified
EsmTherm	K50_dG_Dataset1_Dataset2 domains; 414 wildtypes contribute train mutants under a per-row 80/10/10 split; only 47 domains are test-only	EsmTherm/datasets/split/wildtype_split. csv	protein-level (mutation rows unavailable)
IFUM	Megascale Dataset2: 322 train libraries, 86 in an unseparated val/test block	IFUM published dataset_splits.pdf	protein-level
DDGemb	S2450 = S2648 − 18 sequences/198 mutations >25 % identity to S669; 2,450 mutations / 115 UniProt	ddgemb.biocomp.unibo.it S2450.tsv/.fasta	verified
DDMut	S9028 = S2648 + FireProtDB, 4,514 forward + 4,514 reverse; multi-mutant branch SM1242	biosig.lab.uq.edu.au S9028.csv, SM1242.csv	verified
GeoDDG-3D	S8754 = 8,754 single mutations / 301 proteins (ProThermDB + ThermoMutDB)	GeoStab S8754.csv	verified
ESM3ΔG	960,216-sequence MGnify Stability fine-tune (525,127 WT + 435,089 point mutants); Megascale is its held-out test	Zenodo 19411306 mgnify_training_index.csv	verified (0 exact sequences)
ProSST	18.8 M AlphaFoldDB structures (masked-LM); no ΔΔG fine-tune	ProSST paper	documented only
ProteinMPNN	PDB assemblies deposited before 2021-08-02; no mutation-level supervision	ProteinMPNN paper	documented only
ESM3 (small/medium/large, and the σ = 0 control)	UniRef/MGnify/JGI + PDB pre-2020-05-01 + AFDB v4	ESM3 paper	documented only
Rosetta Cartesian ΔΔG	no training data	physics-based	

Table 6.5: what the audit finds

finding	size	affected rows
23 ProteinGym assays are fully inside the capped train split	23/213 DMS, all Stability	45,623/2,446,730 rows (100 % of those 23 DMS)
ME stage 1/2 and ThermoMPNN on those assays	singles (24,252 exact) + constituent (21,331) for ME; 24,252 exact for TMPNN	same 23 DMS
Domainome domain O95155_PF04564_1229 (UBE4B) is Megascale library 3L1X	1 of 522 domains	1,134 exact + 3,177 same-protein rows (0.8 % of Domainome)
EsmTherm's 80/10/10 is per row, not per domain: 27/33 test libraries contribute training mutants	82.2 % of test rows same-protein	P, M? expected ≈ 0.8
IFUM trained on 21/33 test libraries	45.2 % of test rows same-protein	P, M? = same rows
ESM3ΔG is zero-shot on Megascale	0 exact sequences anywhere in the benchmark	40.7 % strong + 50.3 % weak homologs
DDGemb = S2450	92.5 % of S2648; 30 % of S8754; 28 % of Ssym; 45 % of Q3421; 26 % of K3822; 24 % of K2369	upper bounds via S2648
DDMut = S9028 + SM1242	88.8 % of S2648; 60.3 % of S8754; 42.1 % of S783; 19.6 % of S571; 29.7 % of Ssym; 58.9 % of PTMUL; 36.2 % of PTMUL-D	exact
GeoDDG = S8754	S8754 is its training set (#); 97.7 % of S2648; 88.3 % of S783; 38.5 % of S571; 68.7 % of Q3421; 69.3 % of K3822; 76.1 % of K2369	exact
the "six clean sets" claim	only S669 and S461 have zero identical mutations for every model	DDMut shares 11 %/16 % of them at protein level, GeoDDG 1 %/3 %